Proteograph

Typst proteomics package

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https://codeberg.org/olangella/proteograph

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Abstract

proteograph is a package defining Typst functions dedicated to proteomics.

Contents

Reference

XIC (eXtracted Ion Chromatogram) plot

Example :

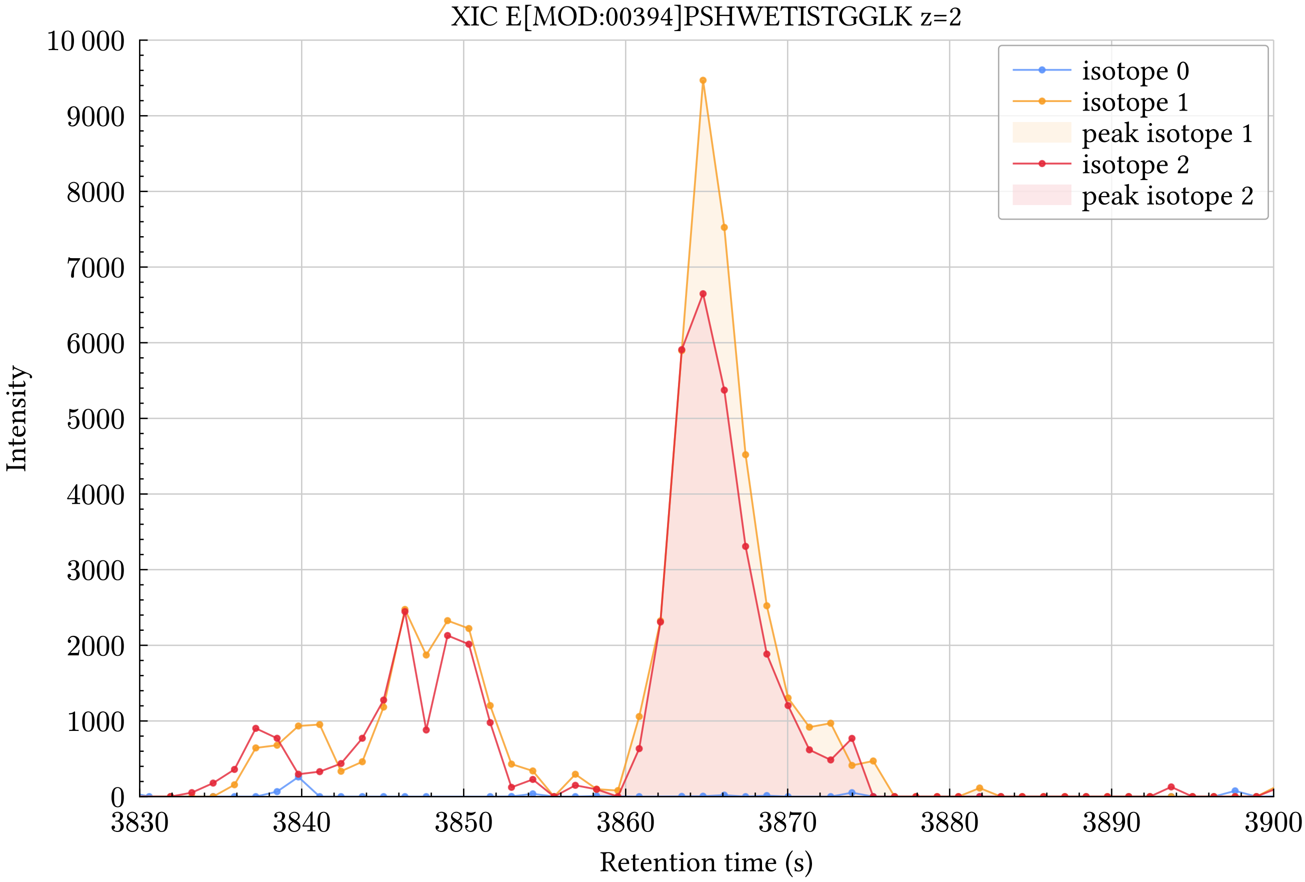
```
#import "@preview/proteograph:0.2.3": *

#let data_json = json("../examples/data/pepa1b35.json")

#let xic0 = data_json.found_list_first_pass.first().xics.first().trace
#xic0.insert("title", "isotope 0")
#let xic1 = data_json.found_list_first_pass.first().xics.at(1).trace
#xic1.insert("title", "isotope 1")
#xic1.insert("peak-
begin", data_json.found_list_first_pass.first().xics.at(1).peak.rt.first())
#xic1.insert("peak-
end", data_json.found_list_first_pass.first().xics.at(1).peak.rt.last())
#let xic2 = data_json.found_list_first_pass.first().xics.at(2).trace
#xic2.insert("title", "isotope 2")
#xic2.insert("peak-
begin", data_json.found_list_first_pass.first().xics.at(2).peak.rt.first())
#xic2.insert("peak-
end", data_json.found_list_first_pass.first().xics.at(2).peak.rt.last())

#xic-plot(
  height: 10cm,
  title: "XIC E[MOD:00394]PSHWETISTGGLK z=2",
  rt-range: (3830, 3900),
  max-intensity: 10000,
  xic0,
  xic1,
  xic2,
)
```

Result :



XIC plot full documentation

- xic-isotope-plot()
- xic-plot()

xic-isotope-plot

Generates a XIC plot separating istope pattern by charge

Parameters

```
xic-isotope-plot(
  width: length relative,
  height: length relative,
  rt-range: auto array,
  max-intensity: none float,
  title: content,
  ..xic-item: dictionary
) -> content
```

width `length` or `relative`

The width of the diagram. This can be

- A `length`; in this case, it defines just the width of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the width of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded width, e.g., a page with `width: auto`.

Default: `15cm`

height `length` or `relative`

The height of the diagram. This can be

- A `length`; in this case, it defines just the height of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the height of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded height, e.g., a page with `height: auto`.

Default: `10cm`

rt-range `auto` or `array`

Data limits along the x-axis (retention time in seconds). Expects auto or a tuple (min, max) where min and max may individually be auto

Default: `auto`

max-intensity `none` or `float`

Maximum intensity to display. **Optional**.

Example: `30000`

Default: `none`

title `content`

Graph title

Default: `none`

..xic-item `dictionary`

dictionary containing XIC data

```typc
/// XIC dictionary structure
#let xic_item = (
  /// Array of retention times values
  "x": (),
  /// Array of intensity values
  "y": (),
  /// Title for this XIC
  "title": "title for this XIC"
  /// Detected peak retention time start. *Optional*.
  "peak-begin": 3500,
  /// Detected peak retention time end. *Optional*.
  "peak-end": 3500,
  /// Charge.
  "charge": none,
  /// Isotope number.
  "isotope": none,
)
```

xic-plot

Generates a XIC plot.

Parameters

```
xic-plot(
  width: length relative,
  height: length relative,
  rt-range: auto array,
  max-intensity: none float,
  title: content,
  ..xic-item: dictionary
) -> content
```

width `length` or `relative`

The width of the diagram. This can be

- A `length`; in this case, it defines just the width of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the width of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded width, e.g., a page with `width: auto`.

Default: `15cm`

height `length` or `relative`

The height of the diagram. This can be

- A `length`; in this case, it defines just the height of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the height of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded height, e.g., a page with `height: auto`.

Default: `10cm`

rt-range `auto` or `array`

Data limits along the x-axis (retention time in seconds). Expects auto or a tuple (min, max) where min and max may individually be auto

Default: `auto`

max-intensity `none` or `float`

Maximum intensity to display. **Optional**.

Example: `30000`

Default: `none`

title `content`

Graph title

Default: `none`

..xic-item `dictionary`

dictionary containing XIC data

```typc
/// XIC dictionary structure
#let xic_item = (
  /// Array of retention times values
  "x": (),
  /// Array of intensity values
  "y": (),
  /// Title for this XIC
  "title": "title for this XIC"
  /// Detected peak retention time start. *Optional*.
  "peak-begin": 3500,
  /// Detected peak retention time end. *Optional*.
  "peak-end": 3500,
)
```

Retention time alignment

The alignment of retention times between MS runs done with MassChroQ3 are represented as follow in the JSON output

MS run alignment data

```
{
    "corrections": {
        "msruna1": {
            "aligned": [],
            "ms2_delta_rt": {
                "x": [],
                "y": []
            },
            "ms2_mean": [],
            "ms2_median": [],
            "original": []
        },
        "msruna2": {
            "aligned": [],
            "ms2_delta_rt": {
                "x": [],
                "y": []
            },
            "ms2_mean": [],
            "ms2_median": [],
            "original": []
        },
        "msruna3": {
            "aligned": [],
            "ms2_delta_rt": {
                "x": [],
                "y": []
            },
            "ms2_mean": [],
            "ms2_median": [],
            "original": []
        },
    },
    "msrun_ref": "msruna4"
}
```

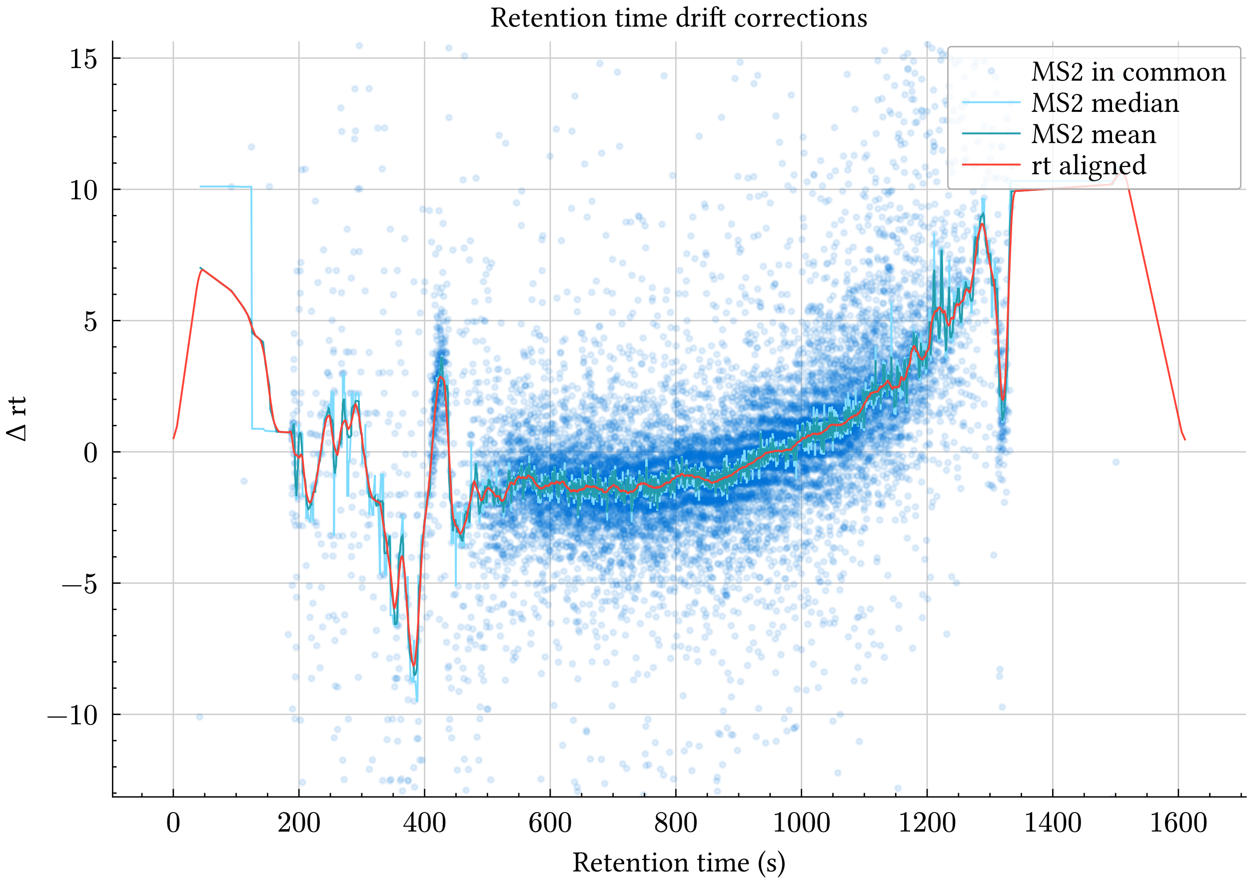
Example :

```
#import "@preview/proteograph:0.2.3": *

#let rt_align = json("../examples/data/one_alignment.json")
#rtalign-plot(
  title: "Retention time drift corrections",
  ylim: auto,
  ms2-delta-rt: rt_align.ms2_delta_rt,
  aligned: rt_align.aligned,
  ms2-mean: rt_align.ms2_mean,
```

```
  ms2-median: rt_align.ms2_median,
  original: rt_align.original,
)
```

Result :



Retention time alignment full documentation

- rtalign-plot()

rtalign-plot

retention time delta plot between MS runs plot.

Parameters

```
rtalign-plot(
  width: length relative,
  height: length relative,
  xlim: auto array,
  ylim: auto array,
  title,
  original,
  aligned,
  ms2-delta-rt,
  ms2-mean,
  ms2-median
) -> content
```

width `length` or `relative`

The width of the diagram. This can be

- A `length`; in this case, it defines just the width of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the width of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded width, e.g., a page with `width: auto`.

Default: `15cm`

height `length` or `relative`

The height of the diagram. This can be

- A `length`; in this case, it defines just the height of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the height of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded height, e.g., a page with `height: auto`.

Default: `10cm`

xlim `auto` or `array`

Data limits along the x-axis (retention time in seconds). Expects auto or a tuple (min, max) where min and max may individually be auto

Default: `auto`

ylim `auto` or `array`

Data limits along the y-axis (retention time delta in seconds). Expects auto or a tuple (min, max) where min and max may individually be auto

Default: `auto`

MS2 spectra

Simple MS2 spectra

The spectra is a dictionnary containing 2 arrays of floats : mz and intensity

spectra dictionnary

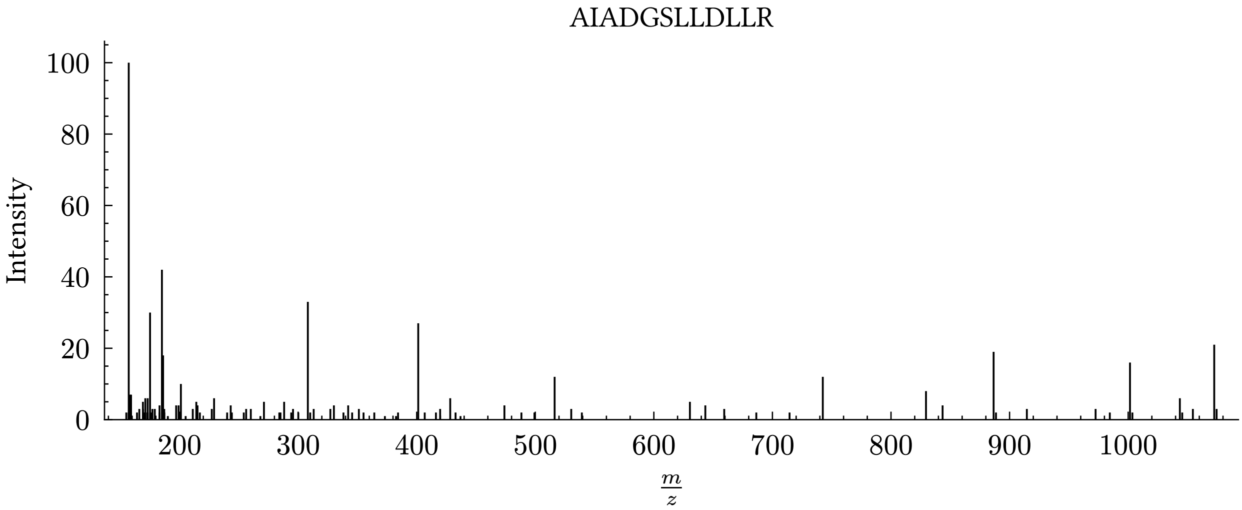
```
{
        "intensity": [
            2,
            2,
            100,
            5,
            7,
            2
        ],
        "mz": [
            155.081,
            157.097,
            157.133,
            158.092,
            158.137,
            159.076
        ]
}
```

Example :

```
#import "@preview/proteograph:0.2.3": *

#let complete_psm = json("../examples/data/complete_psm.json")
#ms2spectra-plot(height: 5cm, title: "AIADGSLLDLLR", spectra: complete_psm.spectra)
```

Result :



MS2 spectra with ion annotations

This example shows an MS2 spectra with ion series annotation

The ion annotations are described in a dictionnary :

ion-series dictionnary

```
{
        "a": [
            {
                "charge": 1,
                "intensity": 100,
                "mz": 157.133,
                "mzth": 157.13353961189497,
                "size": 2
            }
        ],
        "b": [
            {
                "charge": 1,
                "intensity": 42,
                "mz": 185.128,
                "mzth": 185.12845423145498,
                "size": 2
            },
            {
                "charge": 1,
                "intensity": 3,
                "mz": 256.165,
                "mzth": 256.16556801702,
                "size": 3
            }
        ]
}
```

The first key is the ion series name as follow :

y, b, yp, ystar, y*, yO, x, bstar, b*, bO, a, astar, aO, c, z, bp

and any ion can be described by :

- ion-description()

ion-description

Ion description

Parameters

```
ion-description(
   "charge": integer,
   "intensity": float,
   "mz": float,
   "mzth": float,
   "size": integer
)
```

"charge" `integer`

Ion charge

Default: `1`

"intensity" `float`

Ion intensity

Default: `none`

"mz" `float`

Ion experimental mass on charge ratio

Default: `none`

"mzth" `float`

Ion theoretical mass on charge ratio. **Optional**.

Default: `none`

"size" `integer`

Ion size : number of residues for this fragment

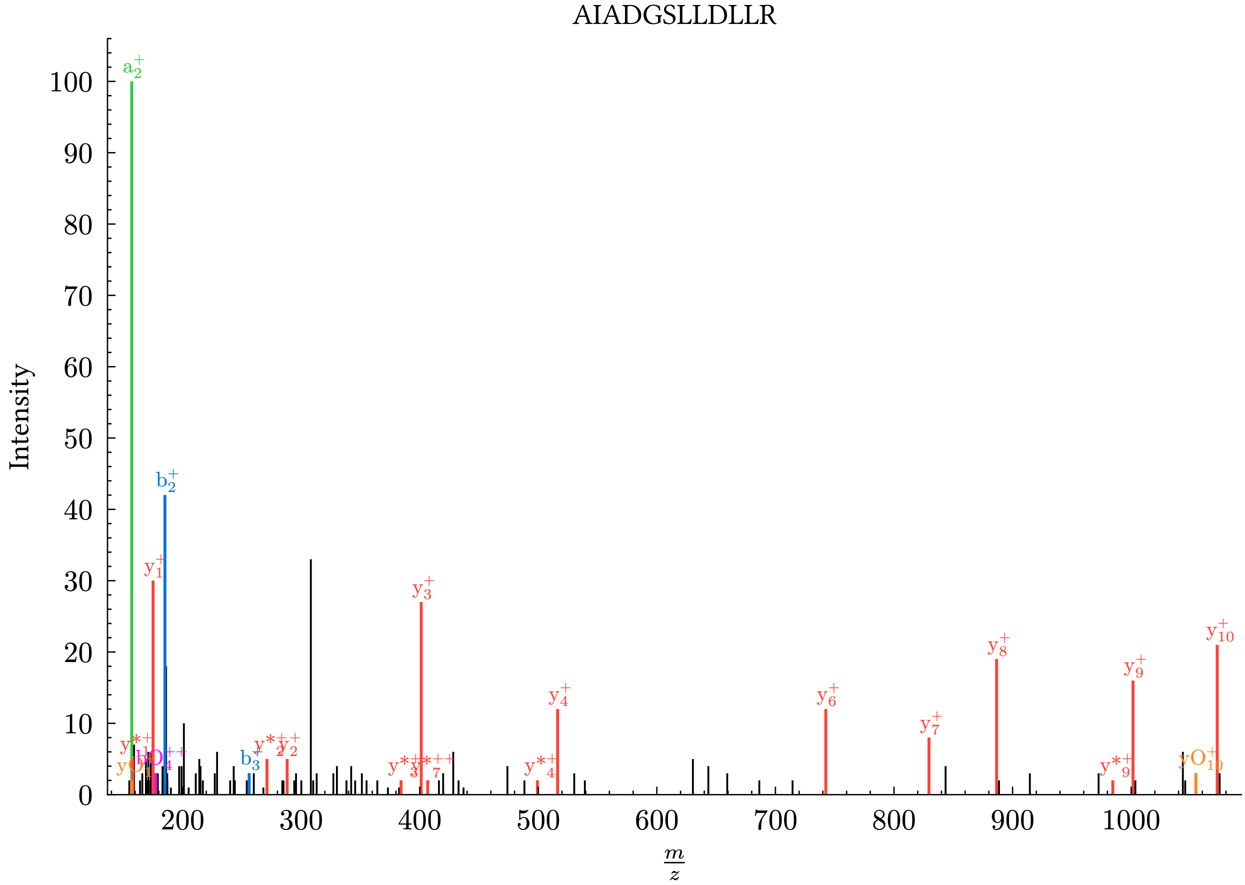
Default: `1`

Example :

```
#import "@preview/proteograph:0.2.3": *

#let complete_psm = json("../examples/data/complete_psm.json")
#ms2spectra-plot(title:  "AIADGSLLDLLR",  spectra:  complete_psm.spectra,  ion-series:
complete_psm.ion-series)
```

Result :



MS2 spectra zoom (mz_range and intensity)

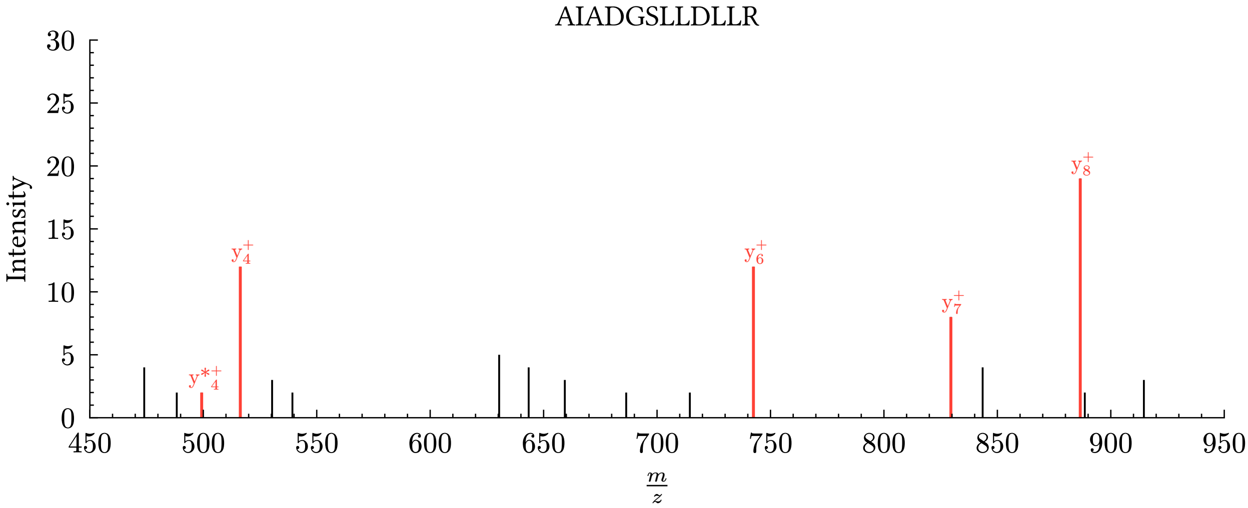
This example shows an MS2 spectra zoomed on a specific m/z range and maximum intensity.

Example :

```
#import "@preview/proteograph:0.2.3": *

#let complete_psm = json("../examples/data/complete_psm.json")
#ms2spectra-plot(
  width: 15cm,
  height: 5cm,
  title: "AIADGSLLDLLR",
  spectra: complete_psm.spectra,
  ion-series: complete_psm.ion-series,
  mz-range: (450, 950),
  max-intensity: 30,
)
```

Result :



MS2 spectra with ion annotations and MS2 fragments mass delta

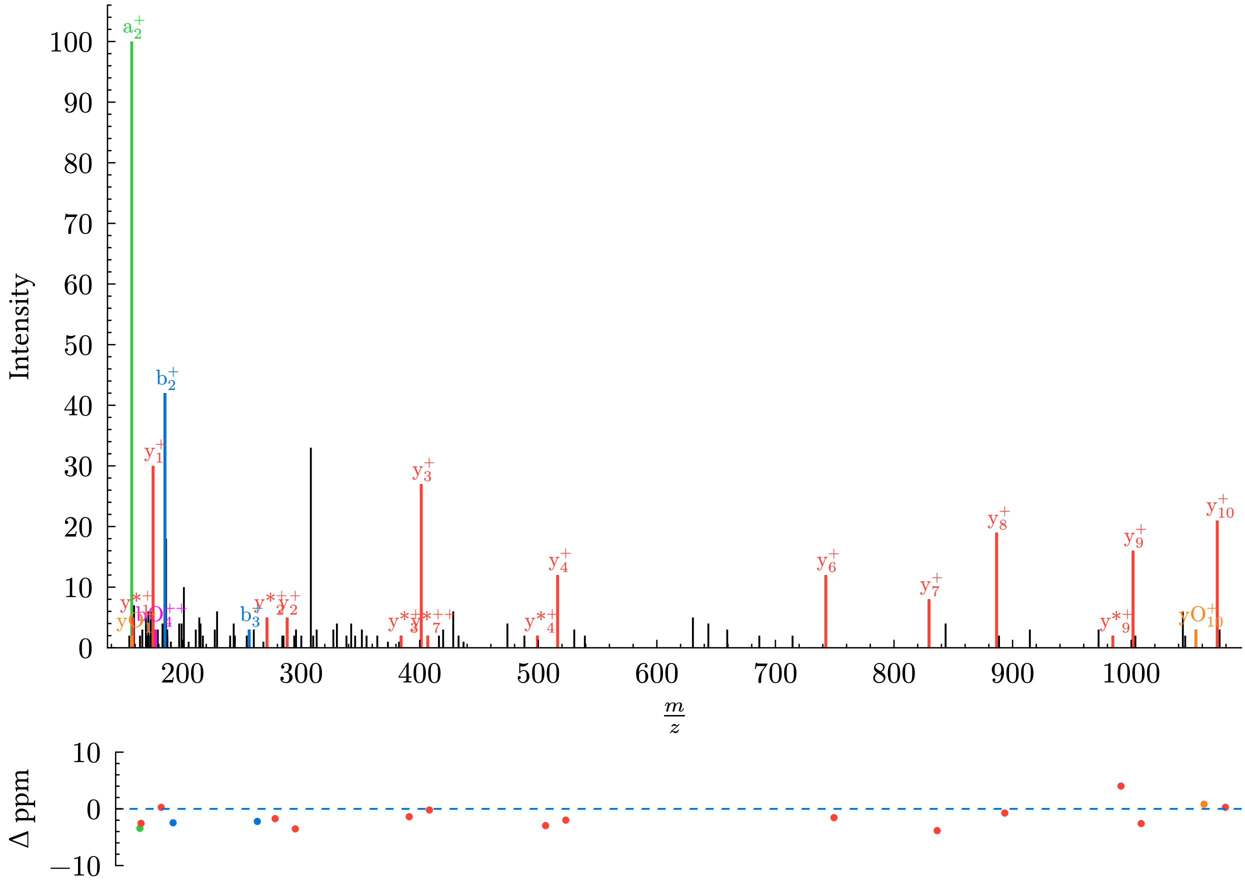
This example shows an MS2 spectra with ion series annotation

Example :

```
#import "@preview/proteograph:0.2.3": *

#let complete_psm = json("../examples/data/complete_psm.json")
#ms2spectra-plot(
  width: 15cm,
  height: 10cm,
  spectra: complete_psm.spectra,
  ion-series: complete_psm.ion-series,
  delta-fragments: true,
)
```

Result :



MS2 spectra with ion annotations and highlighted mass delta between peaks

The space between two peaks can be highlighted and manually annotated using an array defined as follow

Highlight space between peaks array

```
[
{
    "ion": "y",
    "mz": [401.287,516.313],
    "level": 1,
    "label": "D"
},
{
    "ion": "y",
    "mz": [742.481,829.511],
    "level": 1,
    "label": "S"
},
{
    "ion": "y",
    "mz": [829.511,886.535],
    "level": 1,
    "label": "G"
},
```

```
{
    "ion": "y",
    "mz": [886.535,1001.56],
    "level": 1,
    "label": "D"
}
]
```

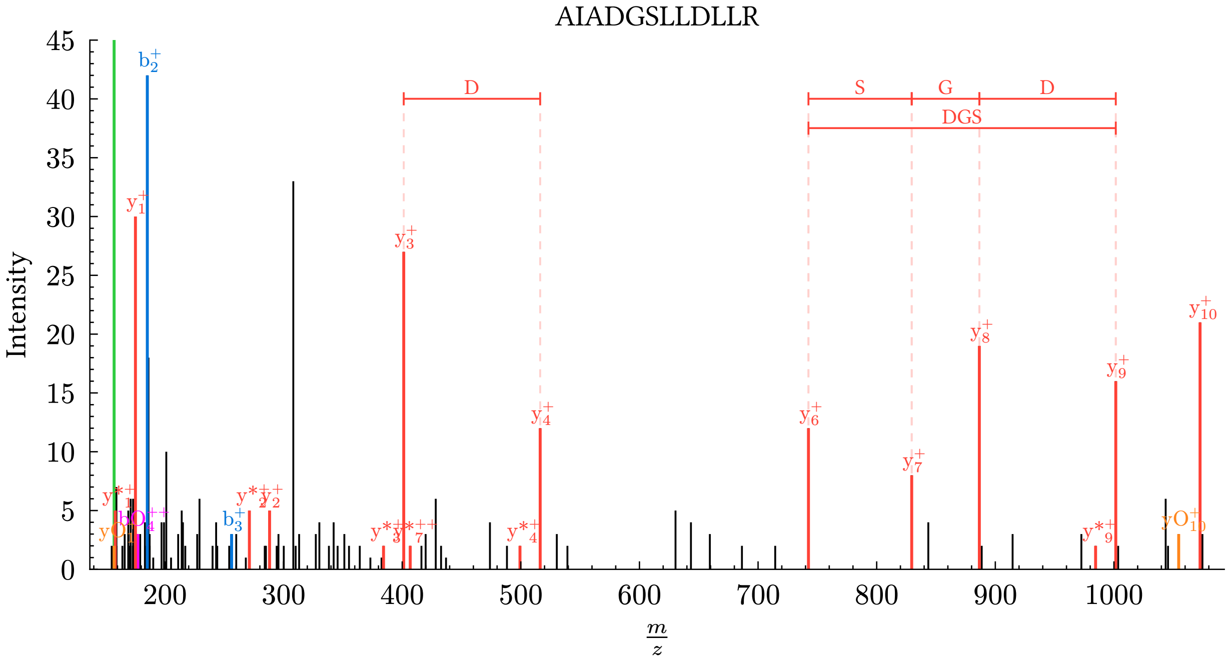
Example :

```
#import "@preview/proteograph:0.2.3": *


#let delta_arr = (
  (
    "ion": "y",
    "mz": (401.287, 516.313),
    "level": 1,
    "label": "D",
  ),
  (
    "ion": "y",
    "mz": (742.481, 829.511),
    "level": 1,
    "label": "S",
  ),
  (
    "ion": "y",
    "mz": (829.511, 886.535),
    "level": 1,
    "label": "G",
  ),
  (
    "ion": "y",
    "mz": (886.535, 1001.56),
    "level": 1,
    "label": "D",
  ),
  (
    "ion": "y",
    "mz": (742.481, 1001.56),
    "level": 2,
    "label": "DGS",
  ),
)

#let complete_psm = json("../examples/data/complete_psm.json")
#ms2spectra-plot(
  width: 15cm,
  height: 7cm,
  max-intensity: 45,
  title: "AIADGSLLDLLR",
  spectra: complete_psm.spectra,
  ion-series: complete_psm.ion-series,
  delta: delta_arr,
)
```

Result :



MS2 spectra full documentation

- ms2spectra-plot()

ms2spectra-plot

Generates an annotated MS2 spectra plot.

Parameters

```
ms2spectra-plot(
  width: length relative,
  height: length relative,
  title: str content,
  mz-range: none array,
  max-intensity: none float,
  spectra: none dictionary,
  ion-series,
  delta,
  delta-fragments: bool
) -> content
```

width `length` or `relative`

The width of the diagram. This can be

- A `length`; in this case, it defines just the width of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the width of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded width, e.g., a page with `width: auto`.

Default: `15cm`

height `length` or `relative`

The height of the diagram. This can be

- A `length`; in this case, it defines just the height of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the height of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded height, e.g., a page with `height: auto`.

Default: `10cm`

title `str` or `content`

Shows the plot title. **Optional**.

Default: `none`

mz-range `none` or `array`

m/z range to display. **Optional**.

Example: `(450, 950)`

Default: `none`

max-intensity `none` or `float`

maximum intensity to display. **Optional**.

Example: `30000`

Default: `none`

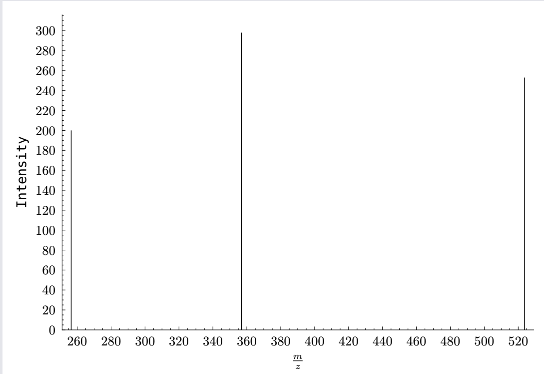
spectra `none` or `dictionary`

Mass spectra values. **Optional**.

a dictionary with the keys `mz` (array of mass to charge ratios) and `intensity` (array of intensities) with the same length for each array

(e.g., `spectra: (mz: (256.45, 356.89, 523.78), intensity: (200, 298 ,253))`)

```typ
#import "@preview/proteograph:0.2.3": *
#set text(size: 12pt)
#ms2spectra-plot( spectra: (mz: (256.45, 356.89, 523.78), intensity: (200, 298, 253)))
```



Default: `none`

delta-fragments `bool`

Whether to clip the matched ion masss delta to the plot. **Optional**.

Default: `false`

Protein sequence

Example :

```
#import "@preview/proteograph:0.2.3": *

#protein-sequence(
  line-length: 60,
 "MASTKAPGPGEKHHSIDAQLRQLVPGKVSEDDKLIEYDALLVDRFLNILQDLHGPSLREFVQECYEVSADYEGKGDTTKLGELGAKLTGLAPADAIL
)
```

Result :

```
  0   MASTKAPGPGEKHHSIDAQLRQLVPGKVSEDDKLIEYDALLVDRFLNILQDLHGPSLREF
 60   VQECYEVSADYEGKGDTTKLGELGAKLTGLAPADAILVASSILHMLNLANLAEEVQIAHR
120   RRNSKLKKGGFADEGSATTESDIEETLKRLVSEVGKSPEEVFEALKNQTVDLVFTAHPTQ
180   SARRSLLQKNARIRNCLTQLNAKDITDDDKQELDEALQREIQAAFRTDEIRRAQPTPQDE
240   MRYGMSYIHETVWKGVPKFLRRVDTALKNIGINERLPYNVSLIRFSSWMGGDRDGNPRVT
300   PEVTRDVCLLARMMAANLYIDQIEELMFELSMWRCNDELRVRAEELHSSSGSKVTKYYIE
360   FWKQIPPNEPYRVILGHVRDKLYNTRERARHLLASGVSEISAESSFTSIEEFLEPLELCY
420   KSLCDCGDKAIADGSLLDLLRQVFTFGLSLVKLDIRQESERHTDVIDAITTHLGIGSYRE
480   WPEDKRQEWLLSELRGKRPLLPPDLPQTDEIADVIGAFHVLAELPPDSFGPYIISMATAP
540   SDVLAVELLQRECGVRQPLPVVPLFERLADLQSAPASVERLFSVDWYMDRIKGKQQVMVG
600   YSDSGKDAGRLSAAWQLYRAQEEMAQVAKRYGVKLTLFHGRGGTVGRGGGPTHLAILSQP
660   PDTINGSIRVTVQGEVIEFCFGEEHLCFQTLQRFTAATLEHGMHPPVSPKPEWRKLMDEM
720   AVVATEEYRSVVVKEARFVEYFRSATPETEYGRMNIGSRPAKRRPGGGITTLRAIPWIFS
780   WTQTRFHLPVWLGVGAAFKFAIDKDVRNFQVLKEMYNEWPFFRVTLDLLEMVFAKGDPGI
840   AGLYDELLVAEELKPFGKQLRDKYVETQQLLLQIAGHKDILEGDPFLKQGLVLRNPYITT
900   LNVFQAYTLKRIRDPNFKVTPQPPLSKEFADENKPAGLVKLNPASEYPPGLEDTLILTMK
960   GIAAGMQNTGA
```

Protein sequence documentation

- protein-sequence()

protein-sequence

Display a protein sequence in a block with amino acid numbering

Parameters

```
protein-sequence(
  line-length,
  protein
) -> content
```

line-length

The maximum length of amino acid to display in a line. **Optional**.

Default: `50`

protein

Amino acid sequence (one letter code) of the protein to display

Protein diagram

This figure is based on the fletcher package features.

Example :

```
#import "@preview/proteograph:0.2.3": *

#let tint(c) = (stroke: c, fill: rgb(..c.components().slice(0, 3), 5%), inset: 1pt)

#protein-diag(
  line-length: 60,
 "MASTKAPGPGEKHHSIDAQLRQLVPGKVSEDDKLIEYDALLVDRFLNILQDLHGPSLREFVQECYEVSADYEGKGDTTKLGELGAKLTGLAPADAIL
  (pos: (1, 5), style: tint(green)),
  (pos: (12, 17), style: tint(teal)),
  (pos: (75, 177), style: tint(red)),
)
```

Result :

```
MASTKAPGPGEKHHSIDAQLRQLVPGKVSEDDKLIEYDALLVDRFLNILQDLHGPSLREF
VQECYEVSADYEGKGDTTKLGELGAKLTGLAPADAILVASSILHMLNLANLAEEVQIAHR
RRNSKLKKGGFADEGSATTESDIEETLKRLVSEVGKSPEEVFEALKNQTVDLVFTAHPTQ
SARRSLLQKNARIRNCLTQLNAKDITDDDKQELDEALQREIQAAFRTDEIRRAQPTPQDE
MRYGMSYIHETVWKGVPKFLRRVDTALKNIGINERLPYNVSLIRFSSWMGGDRDGNPRVT
PEVTRDVCLLARMMAANLYIDQIEELMFELSMWRCNDELRVRAEELHSSSGSKVTKYYIE
FWKQIPPNEPYRVILGHVRDKLYNTRERARHLLASGVSEISAESSFTSIEEFLEPLELCY
KSLCDCGDKAIADGSLLDLLRQVFTFGLSLVKLDIRQESERHTDVIDAITTHLGIGSYRE
WPEDKRQEWLLSELRGKRPLLPPDLPQTDEIADVIGAFHVLAELPPDSFGPYIISMATAP
SDVLAVELLQRECGVRQPLPVVPLFERLADLQSAPASVERLFSVDWYMDRIKGKQQVMVG
YSDSGKDAGRLSAAWQLYRAQEEMAQVAKRYGVKLTLFHGRGGTVGRGGGPTHLAILSQP
PDTINGSIRVTVQGEVIEFCFGEEHLCFQTLQRFTAATLEHGMHPPVSPKPEWRKLMDEM
AVVATEEYRSVVVKEARFVEYFRSATPETEYGRMNIGSRPAKRRPGGGITTLRAIPWIFS
WTQTRFHLPVWLGVGAAFKFAIDKDVRNFQVLKEMYNEWPFFRVTLDLLEMVFAKGDPGI
AGLYDELLVAEELKPFGKQLRDKYVETQQLLLQIAGHKDILEGDPFLKQGLVLRNPYITT
LNVFQAYTLKRIRDPNFKVTPQPPLSKEFADENKPAGLVKLNPASEYPPGLEDTLILTMK
GIAAGMQNTGA
```

Protein diagram documentation

- protein-diag()

protein-diag

Display a protein sequence in a block with amino acid numbering

Parameters

```
protein-diag(
  line-length: int,
  boxes: array,
  protein: str,
  ..box: dictionary
) -> content
```

line-length `int`

The maximum length of amino acid to display in a line. **Optional**.

Default: `50`

boxes `array`

Array of boxes to draw in the protein sequence **Optional**.

This can be used to visualize the position of one or more peptides

example for 3 peptides :

((pos: (1,5), style: tint(green)),(pos: (12,17), style: tint(teal)),(pos: (75,177), style: tint(red)))

Default: `()`

protein `str`

Amino acid sequence (one letter code) of the protein to display

..box `dictionary`

box item to draw in the protein sequence **Optional**.

This can be used to visualize the position of one or more peptides

example :

(pos: (1,5), style: tint(green))

```typ
#import "@preview/proteograph:0.2.3": *
#protein-diag(line-length: 60,
"MASTKAPGPGEKHHSIDAQLRQLV",
(pos: (1,5), style: (stroke: red, inset: 1pt)),
(pos: (12,17), style: (stroke: teal, inset: 3pt))
)
```

MASTKAPGPGEKHHSIDAQLRQLV

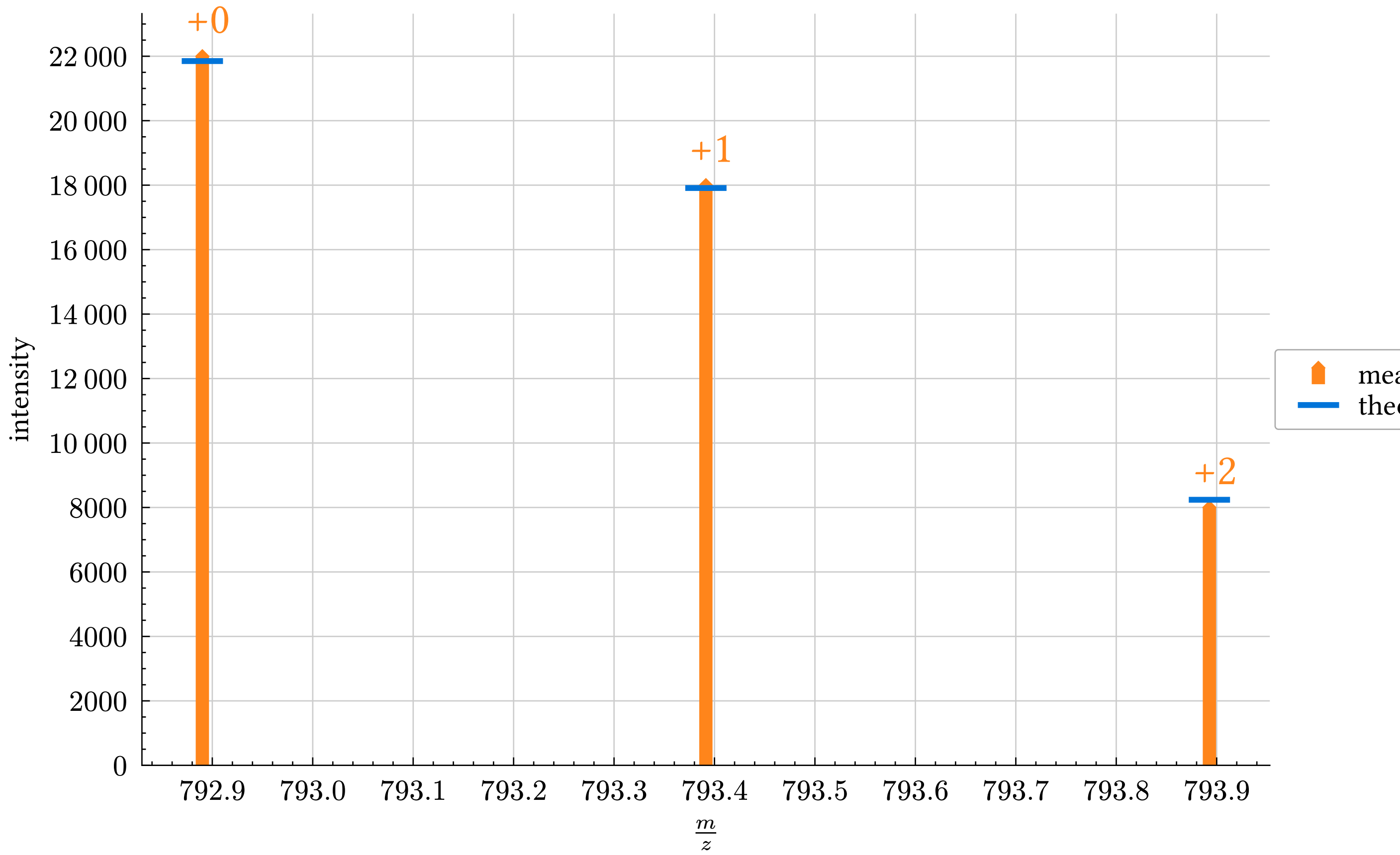
Isotope pattern plot

Example :

```
#import "@preview/proteograph:0.2.3": *


#isotope-pattern-plot(
  mz: (792.89009986061, 793.3914387289032, 793.89274444414951),
  intensity: (22000, 18000, 8000),
  th-ratio: (0.4182709626684006, 0.3428843140088085, 0.157722041765769),
  color: orange,
)
```

Result :



+0
+1
+2
22 000
20 000
18 000
16 000
14 000
12 000
10 000
8000
6000
4000
2000
0
intensity
792.9
793.0
793.1
793.2
793.3
793.4
793.5
793.6
793.7
793.8
793.9
$\frac{m}{z}$
mea
theo